LAB WORKSHEET

Lab 29 — Troubleshooting Common Hardware Failures

Name	______________________________	Date	__________________

Objective: 4.2 — Troubleshoot hardware failures: memory, thermal, POST, CMOS, predictive failures.

Work through the lab on the Killercoda Ubuntu playground and record your results below.

Task 1 — Logs first

Scan dmesg and journal for hardware markers.

Note any mce/edac/temperature entries (or none).

Task 2 — Thermal

Run a stress-ng CPU test and read sensors.

Record temps before/after (or note VM limitation).

Task 3 — SMART

Run smartctl -H and list the predictive attributes.

Name the attributes that predict failure.

Task 4 — POST cues

Match beep/LED/smell cues to causes.

Fill three cue -> cause rows.

Task 5 — VM vs HW

Distinguish a misallocated VM from a hardware fault.

List the three common VM misallocation causes.

Completion checklist

- Investigated event logs before hardware.
- Read SMART predictive attributes.
- Mapped crash-dump terms and POST/LED/beep cues.
- Linked clock skew to CMOS battery failure.
- Separated host faults from VM misallocation.